Comparative Research on GPU Operating Systems and Their Application in AWS Cloud Platforms

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1. Introduction

Modern computing systems have experienced a revolution courtesy of the graphics processing unit (GPU) that has the capacity to process a high amount of data simultaneously. GPUs are initially built to speed up graphics drawing; however, nowadays they are widely used in artificial intelligence, machine learning, scientific modeling, cloud computing, and high-performance computing. Their parallel design is massive, which guarantees them faster performance of compute-intensive workloads when compared to conventional Central Processing Units (CPUs).

Linux OSs are essential in the environment of the use of the GPUs due to their stability, flexibility, and good support on their driver and kernel. Linux GPU operating system command-line tools enable system administrators and researchers to keep track of hardware health, view the behavior of kernel drivers, monitor memory usage, and debug performance or compatibility problems. They are visibly crucial, especially in virtualized and cloud environments where the direct access to the hardware is abstracted.

Business cloud computing services like Amazon Web Services (AWS) and Microsoft Azure allow on-demand services to scalable computing infrastructure, including duration CPU virtual machines. These platforms enable their users to launch applications without any physical investment in physical infrastructure, but with an ultra-thin slice of compute, storage, and networking resources (Migale, 2024). It is vital to know the way in which the tools of the operating system of GPUs are integrated with cloud platforms in order to ensure effective resource utilization and troubleshooting.

The report is an analysis of the practical and comparative use of GPUs using command-line tools that handle the Linux GPU operating system as well as the application of this tool in an AWS cloud infrastructure. Task 1 is dedicated to the identification of the GPUs, their monitoring, and diagnostics with the help of Kali Linux. Task 2 illustrates the process of actual deployment of cloud infrastructure with AWS EC2. Task 3 holds a reflective factor of Linux GPU tools supporting cloud computing, whereas Task 4 acts as a conclusion of the study.

Task 1: Linux GPU Operating Systems Command-Line Tools

2. Overview of Linux GPU Command-Line Tools

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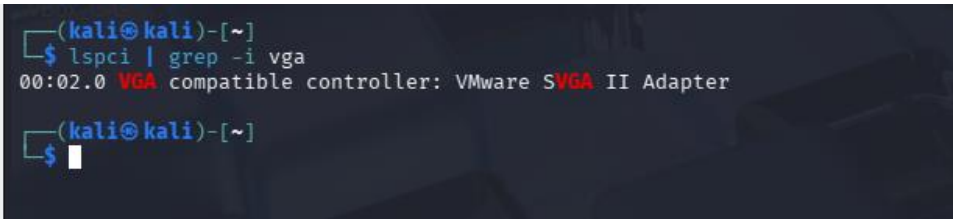

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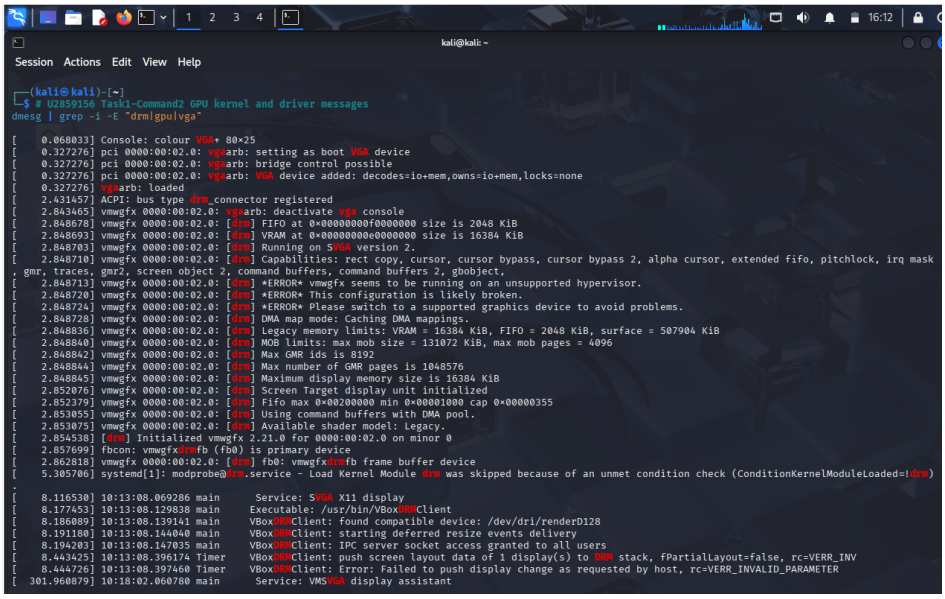

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```
┌──(kali㉿kali)-[~]
└─$ # U2859156 Task1-Command3 GPU driver module status
lsmod | grep -i -E "drm|vga|vmwgfx"

vmwgfx                 491520  2
drm_ttm_helper          16384  2 vmwgfx
ttm                    135168  2 vmwgfx,drm_ttm_helper
drm_client_lib          12288  1 vmwgfx
drm_kms_helper         258048  3 vmwgfx,drm_ttm_helper,drm_client_lib
drm                    831488  8 vmwgfx,drm_kms_helper,drm_ttm_helper,drm_client_lib,ttm

┌──(kali㉿kali)-[~]
└─$
```

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```
┌──(kali㉿kali)-[~]
└─$ # U2859156 Task1-Command4 GPU-related memory overview
free -h

               total        used        free      shared  buff/cache   available
Mem:           1.9Gi       883Mi       715Mi       9.8Mi       521Mi       1.1Gi
Swap:          1.9Gi          0B       1.9Gi

┌──(kali㉿kali)-[~]
└─$
```

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```
Session  Actions  Edit  View  Help
Every 2.0s: cat /proc/fb                                   kali: Sat Jan  3 16:16:01 2026

0 vmwgfxdrmfb
```

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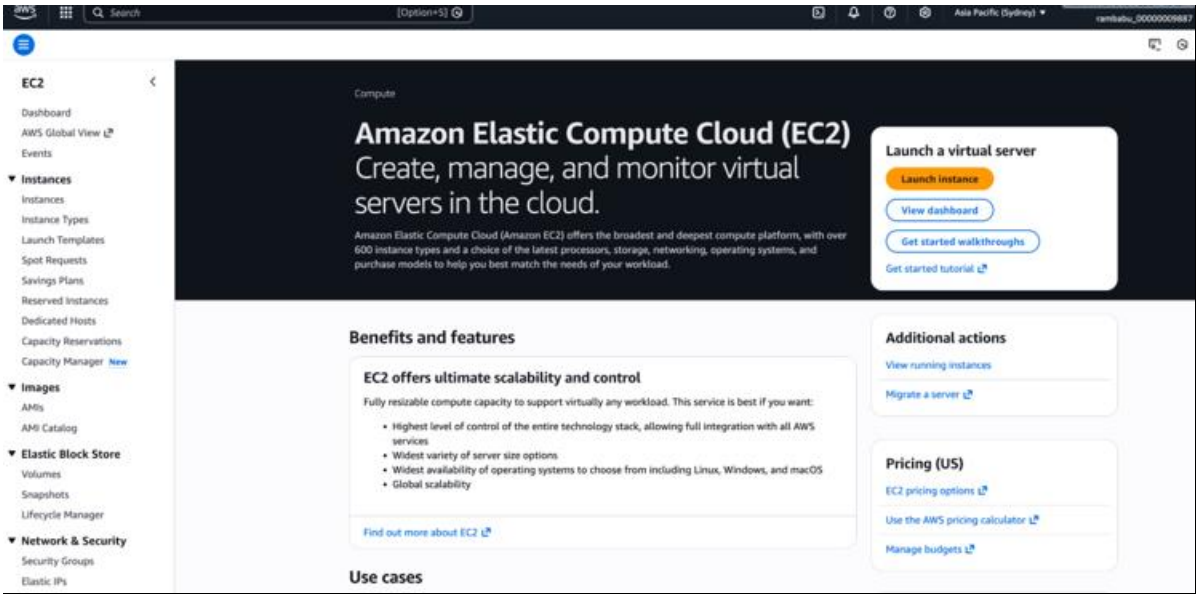

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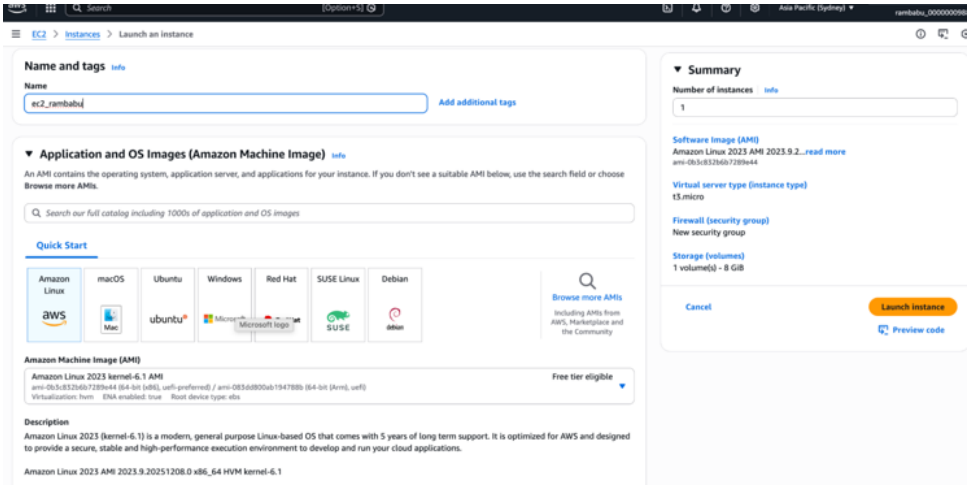
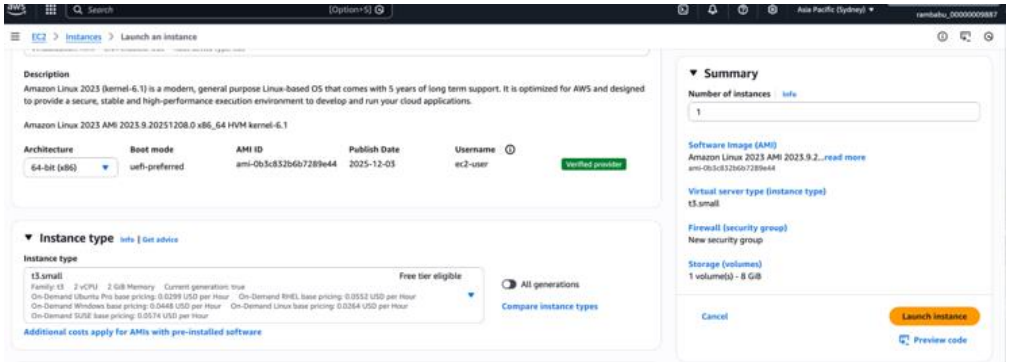

EC2 > Instances > Launch an instance
Name and tags Info
Name
ec2_rambabu
Add additional tags
Application and OS Images (Amazon Machine Image) Info
An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose Browse more AMIs.
Search our full catalog including 1000s of application and OS images
Quick Start
Amazon Linux
macOS
Ubuntu
Windows
Red Hat
SUSE Linux
Debian
Browse more AMIs
Including AMIs from AWS, Marketplace and the Community
Amazon Machine Image (AMI)
Amazon Linux 2023 kernel-6.1 AMI
Free tier eligible
Description
Amazon Linux 2023 (kernel-6.1) is a modern, general purpose Linux-based OS that comes with 5 years of long term support. It is optimized for AWS and designed to provide a secure, stable and high-performance execution environment to develop and run your cloud applications.
Amazon Linux 2023 AMI 2023.9.20251208.0 x86_64 HVM kernel-6.1
Summary
Number of instances Info
1
Software Image (AMI)
Amazon Linux 2023 AMI 2023.9.2...read more
Virtual server type (instance type)
t3.micro
Firewall (security group)
New security group
Storage (volumes)
1 volume(s) - 8 GiB
Cancel
Launch instance
Preview code



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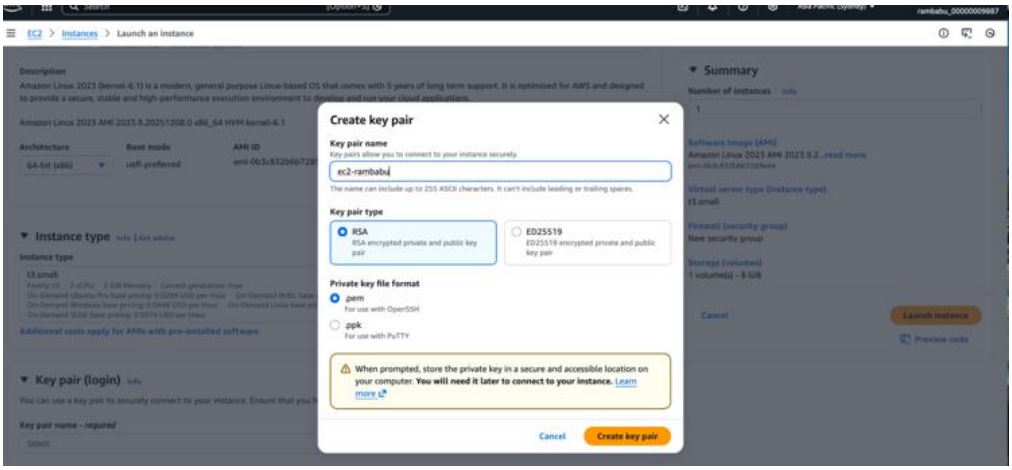

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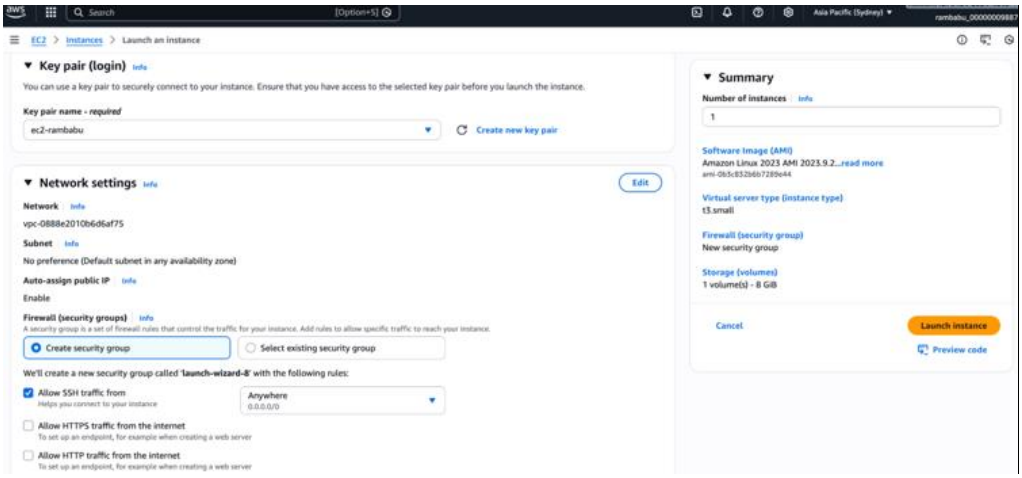

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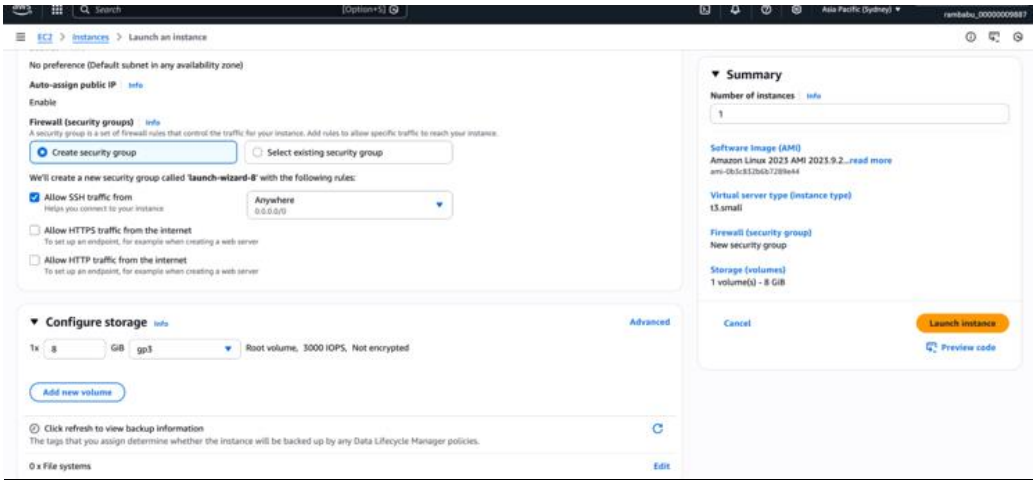

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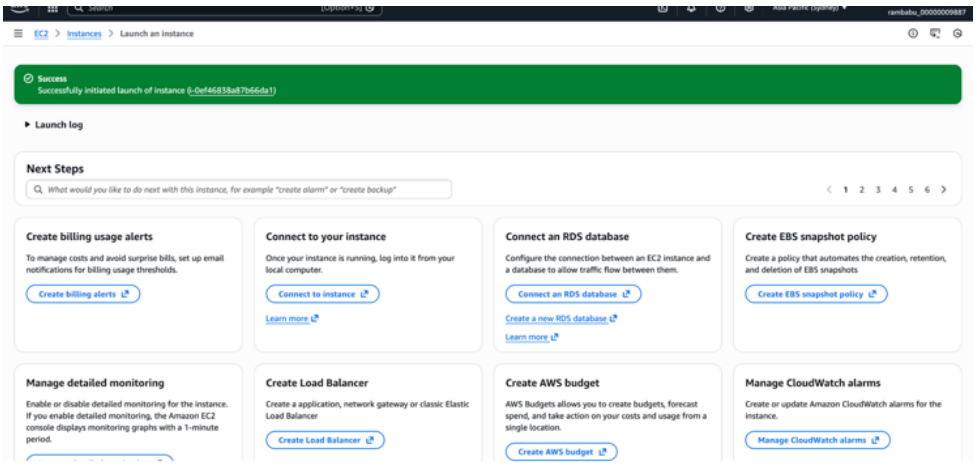

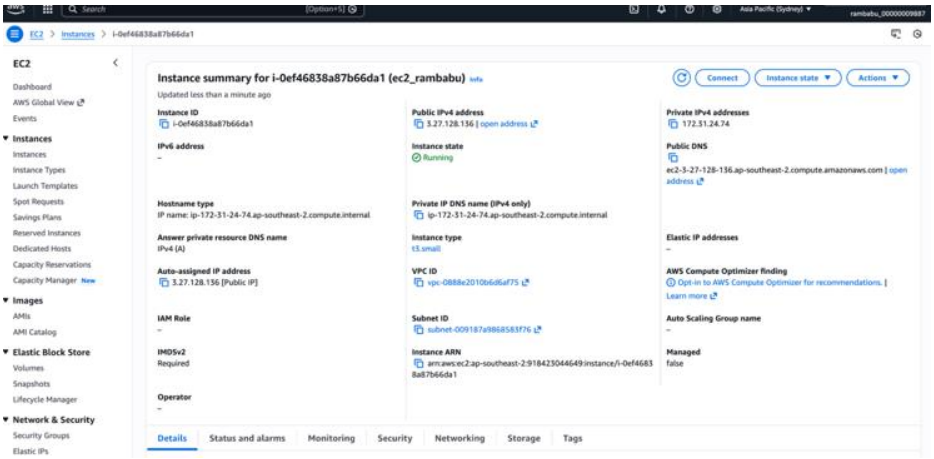

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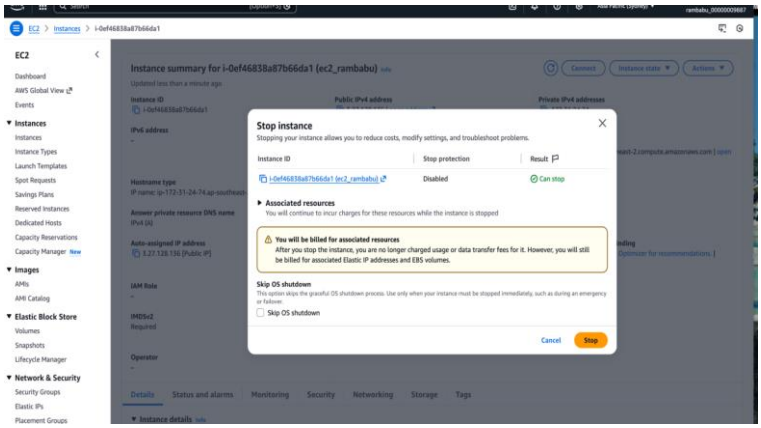

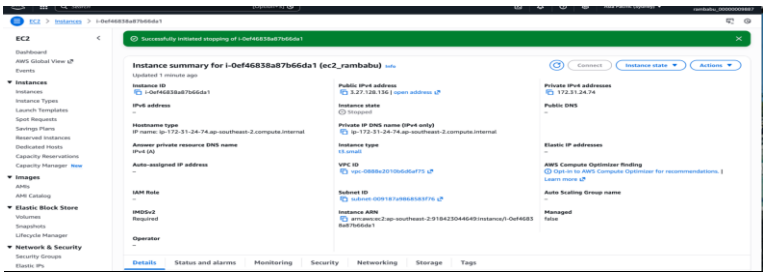

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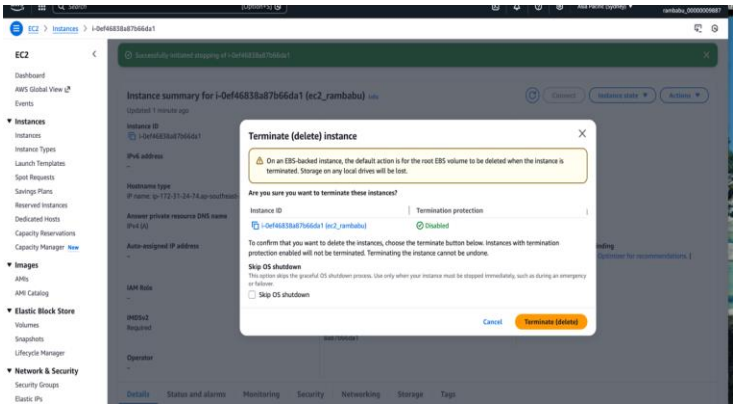

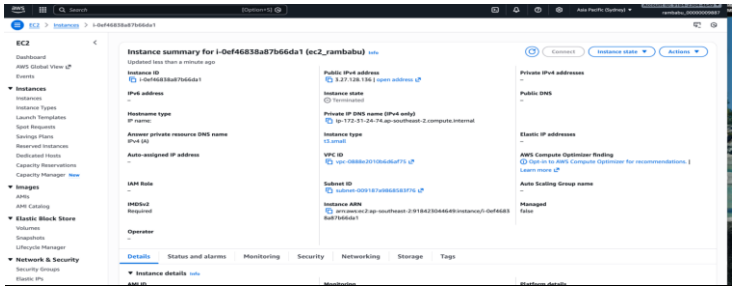

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